

Sanjai S

Biomedical Engineer | AI in Healthcare | Medical Instrumentation Enthusiast | Embedded Systems Developer

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SUMMARY

Biomedical Engineering student passionate about developing intelligent healthcare solutions that combine medical instrumentation, embedded systems, computer vision, and artificial intelligence. Hands-on experience in medical equipment maintenance, prosthetic development, image processing, IoT systems, and AI-powered healthcare applications, with a focus on early diagnosis, patient monitoring, and healthcare accessibility.

EDUCATION

Bachelor of Biomedical Engineering

Currently Pursuing

Higher Secondary Education — Model Higher Secondary School, Nambiyur

2021 – 2023

Secondary Education — SRC M.M. Higher Secondary School, Punjaipuliampatti

2020 – 2021

INTERNSHIP EXPERIENCE

Madras Medical Mission Hospital

June 2025

- Assisted biomedical engineers in equipment maintenance and calibration.
- Observed ECG systems, ventilators, and patient monitors.
- Learned preventive maintenance procedures.
- Gained exposure to clinical engineering practices.
- Developed understanding of biomedical equipment management in hospital environments.

Medsby Healthcare & Engineering Solutions

January 2025

- Learned 3D printing workflow and healthcare applications.
- Fabricated a prosthetic hand prototype.
- Studied prosthetic design methodologies.
- Worked with image processing concepts using OpenCV.
- Observed healthcare technology integration in rehabilitation engineering.

PROJECTS

RetinaAI — Portable AI-Based Ocular Disease Screening Platform

Real-time ocular disease screening from retinal images for accessible, fast diagnostics in remote healthcare environments.

Tech: Deep Learning, Computer Vision, Python, OpenCV

Smart Moss-Based Air Purification & CO₂ Monitoring

IoT-enabled air purification using moss and environmental sensors with real-time CO₂ measurement.

Tech: MQ135 Sensor, Arduino, IoT

Lightweight Deep Learning Model for Small Protein Structure Prediction

Optimized deep learning model predicting small protein structures with reduced computational requirements.

Tech: Deep Learning, Python, AI

TECHNICAL SKILLS

Programming: Java, Python, C Programming

AI & Machine Learning: Deep Learning, Machine Learning, Computer Vision, OpenCV

Embedded Systems & IoT: Arduino, ESP32, Raspberry Pi, Sensor Integration, IoT Systems

Biomedical Engineering: Medical Instrumentation, Biomedical Signal Processing, Healthcare Technology, Medical Device Maintenance

Design & Tools: Blender, 3D Printing, Git & GitHub, Figma